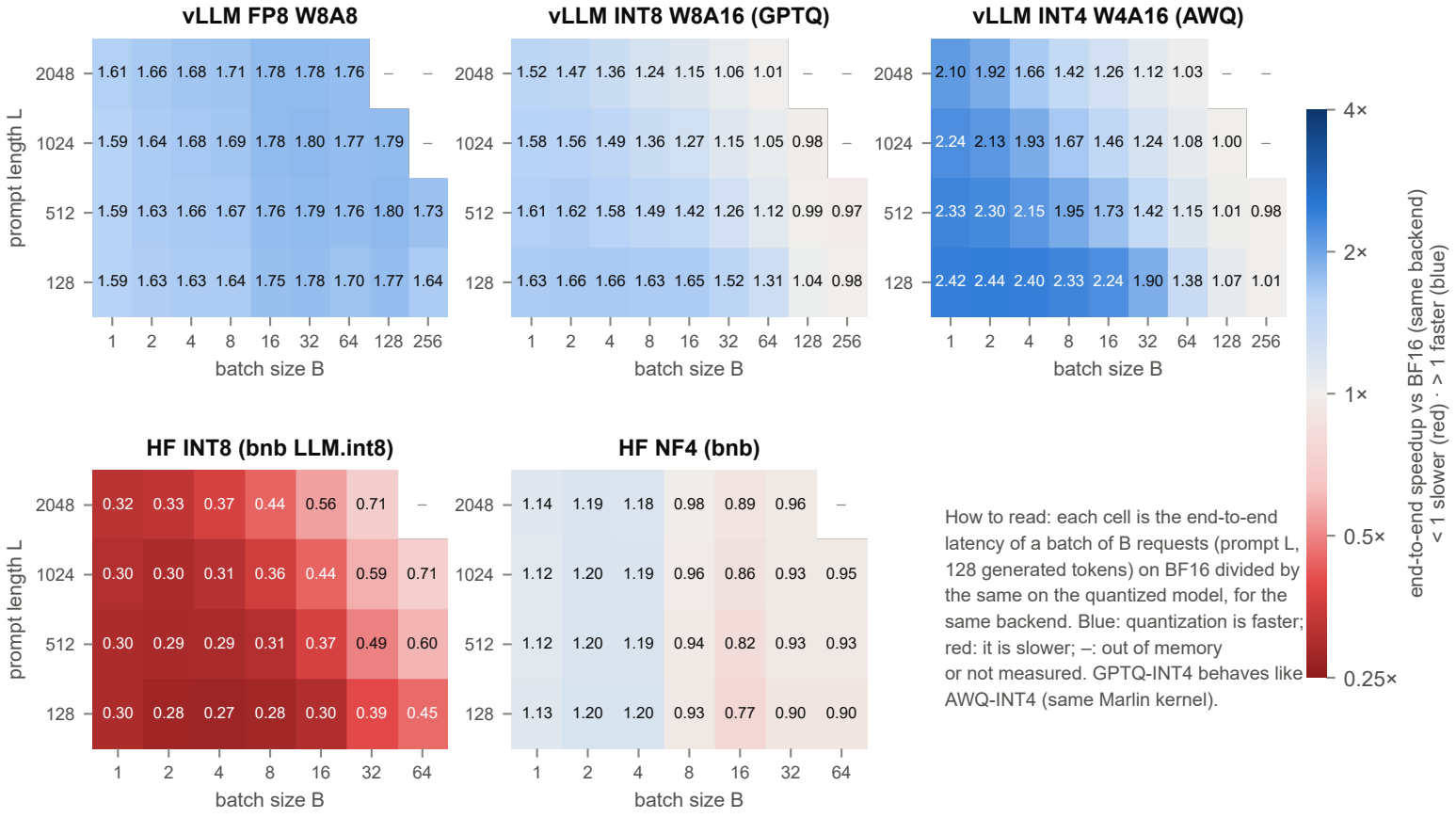

vLLM FP8 W8A8
2048 1.61 1.66 1.68 1.71 1.78 1.78 1.76 – –
1024 1.59 1.64 1.68 1.69 1.78 1.80 1.77 1.79 –
512 1.59 1.63 1.66 1.67 1.76 1.79 1.76 1.80 1.73
128 1.59 1.63 1.63 1.64 1.75 1.78 1.70 1.77 1.64
1 2 4 8 16 32 64 128 256
batch size B
vLLM INT8 W8A16 (GPTQ)
2048 1.52 1.47 1.36 1.24 1.15 1.06 1.01 – –
1024 1.58 1.56 1.49 1.36 1.27 1.15 1.05 0.98 –
512 1.61 1.62 1.58 1.49 1.42 1.26 1.12 0.99 0.97
128 1.63 1.66 1.66 1.63 1.65 1.52 1.31 1.04 0.98
1 2 4 8 16 32 64 128 256
batch size B
vLLM INT4 W4A16 (AWQ)
2048 2.10 1.92 1.66 1.42 1.26 1.12 1.03 – –
1024 2.24 2.13 1.93 1.67 1.46 1.24 1.08 1.00 –
512 2.33 2.30 2.15 1.95 1.73 1.42 1.15 1.01 0.98
128 2.42 2.44 2.40 2.33 2.24 1.90 1.38 1.07 1.01
1 2 4 8 16 32 64 128 256
batch size B
prompt length L
HF INT8 (bnb LLM.int8)
2048 0.32 0.33 0.37 0.44 0.56 0.71 –
1024 0.30 0.30 0.31 0.36 0.44 0.59 0.71
512 0.30 0.29 0.29 0.31 0.37 0.49 0.60
128 0.30 0.28 0.27 0.28 0.30 0.39 0.45
1 2 4 8 16 32 64
batch size B
HF NF4 (bnb)
2048 1.14 1.19 1.18 0.98 0.89 0.96 –
1024 1.12 1.20 1.19 0.96 0.86 0.93 0.95
512 1.12 1.20 1.19 0.94 0.82 0.93 0.93
128 1.13 1.20 1.20 0.93 0.77 0.90 0.90
1 2 4 8 16 32 64
batch size B
prompt length L
4×
2×
1×
0.5×
0.25×
end-to-end speedup vs BF16 (same backend)
< 1 slower (red) · > 1 faster (blue)
How to read: each cell is the end-to-end latency of a batch of B requests (prompt L, 128 generated tokens) on BF16 divided by the same on the quantized model, for the same backend. Blue: quantization is faster; red: it is slower; –: out of memory or not measured. GPTQ-INT4 behaves like AWQ-INT4 (same Marlin kernel).